

Executive Summary: Telecom Customer Churn Analysis

Overview

An analysis was conducted on **7,043 telecom customers** to identify the primary drivers of customer churn and uncover opportunities to improve retention. Data quality issues were addressed by replacing missing **TotalCharges** values with 0 for customers with zero tenure and converting the field to a numeric format. No duplicate records or missing values remained after preprocessing.

Key Business Metric

- **Total Customers:** 7,043
- **Churned Customers:** 26.54%
- **Retained Customers:** 73.46%

This indicates that approximately **1 in every 4 customers leaves the company**, representing a significant opportunity for revenue protection through targeted retention strategies.

Key Findings :

1. New Customers Are Most Likely to Churn

Analysis of customer tenure revealed a strong relationship between customer age and retention.

Insights

- Churn is highest among customers in their **first few months (1–2 months)** of service.
- Customers with longer tenure demonstrate substantially higher retention rates.
- The average customer tenure is approximately **32 months**, but churn is heavily concentrated among recently acquired customers.

Business Impact

The onboarding and early customer experience phase is the most critical period for preventing churn.

Recommended Actions

- Implement a structured **90-day onboarding program**.

- Offer welcome incentives and personalized support during the first three months.
- Monitor new customers closely for service issues and engagement levels.

2. Contract Type Is the Strongest Churn Indicator

Contract duration shows one of the clearest relationships with customer retention.

Insights

- Customers on **month-to-month contracts** exhibit the highest churn rates.
- Customers on **one-year and two-year contracts** are significantly more likely to remain with the company.

Business Impact

Customers without long-term commitments can switch providers more easily, increasing churn risk.

Recommended Actions

- Encourage migration from month-to-month plans to annual contracts.
- Offer loyalty discounts and contract renewal incentives.
- Introduce bundled packages with long-term pricing benefits.

3. Senior Citizens Represent a Higher-Risk Segment

The analysis found that senior citizens churn at a higher rate than non-senior customers.

Insights

- Although senior citizens represent a smaller portion of the customer base, their churn percentage is comparatively higher.
- This may indicate challenges related to pricing, service complexity, or customer support experiences.

Recommended Actions

- Create senior-focused support programs.
- Simplify service plans and billing communication.
- Provide dedicated customer assistance channels.

4. Value-Added Services Improve Retention

Customers subscribing to additional services demonstrate stronger loyalty.

Services Associated with Lower Churn

- Online Security
- Online Backup
- Tech Support
- Device Protection

Insights

Customers who adopt these services are more engaged and derive greater value from the telecom ecosystem.

Business Impact

Higher service adoption creates stronger customer dependence on the platform and reduces switching behavior.

Recommended Actions

- Promote security and support add-ons during onboarding.
- Bundle protection and support services with internet plans.
- Use targeted campaigns for customers who currently have only basic services.

5. Payment Method Is a Significant Churn Predictor

Payment behavior reveals a strong churn pattern.

Insights

- Customers paying via **Electronic Check** exhibit significantly higher churn rates than customers using other payment methods.

Possible Reasons

- Less automated payment process.
- Lower customer commitment.
- Potential dissatisfaction or financial instability.

Recommended Actions

- Encourage migration to auto-pay options.
- Provide incentives for credit card or bank auto-debit enrollment.
- Monitor electronic-check users as a priority churn-risk segment.

6. Gender Has Minimal Influence on Churn

The gender analysis showed no meaningful difference between male and female customer churn behavior.

Business Impact

Gender-specific retention campaigns are unlikely to generate significant improvements.

Recommended Actions

Focus retention efforts on behavioral and service-related factors rather than demographic targeting.

Customer Segments at Highest Risk:

High-Risk Customers

Customers exhibiting multiple characteristics below should be prioritized by retention teams:

- Month-to-month contract holders
- Customers with tenure under 6 months
- Electronic-check payment users
- Senior citizens
- Customers without:
 - Online Security
 - Online Backup
 - Tech Support
 - Device Protection

These customers represent the most vulnerable segment for future churn.

Strategic Recommendations:

Immediate Priorities

1. Reduce early-life churn through onboarding programs.
2. Convert month-to-month customers into annual contracts.
3. Increase adoption of value-added services.
4. Target electronic-check customers with retention campaigns.
5. Develop specialized retention initiatives for senior citizens.

Expected Business Benefits

- Lower customer acquisition replacement costs.
- Higher customer lifetime value (CLV).
- Improved recurring revenue stability.
- Increased profitability through service bundling and longer contract durations.

Executive Conclusion:

The analysis shows that churn is primarily driven by **short customer tenure, month-to-month contracts, lack of value-added services, senior-citizen status, and electronic-check payment methods**. With **26.54% of customers churning**, the company has a substantial opportunity to improve retention by focusing on customer onboarding, contract conversion, service bundling, and proactive engagement of high-risk segments. Targeting these areas is likely to produce the greatest reduction in churn and the strongest impact on long-term revenue growth.